

SOCIAL NETWORKS

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Week 8 — Question Paper (Comprehension-Based MCQ / MSQ)

Comprehension 1 — Playlists and Songs: Hubs and Authorities

Case Study

An engineer at a music-streaming service wants to score both songs and the playlists that feature them. The setup is a directed graph: a playlist points to each song it features. Two roles emerge. A playlist acts as a hub a collection that points outward to content while a song acts as an authority something that is pointed at. Intuitively, a good playlist is one that features good songs, and a good song is one that is featured by good playlists. The definition is therefore recursive, and the engineer scores it by the principle of repeated improvement.

The procedure assigns every node two numbers and updates them together. In each round, a hub's score is set to the sum of the authority scores of the songs it points to, and an authority's score is set to the sum of the hub scores of the playlists that point to it; after each round both sets of scores are normalized. The engineer notes that in a fully general directed graph this is not limited to a neat split into “only playlists” and “only songs”: every node carries both kinds of score at the same time.

Running the update repeatedly, the engineer observes that the scores settle: the relative proportions between nodes stop changing, even though the raw numbers are rescaled each round. This convergence is guaranteed for any graph one always reaches a stable point. The same idea once underpinned the rating of “news lists” in the late 1990s, where a list was trusted to the extent that it pointed to genuinely good articles.

Questions 1–10

Q1. [MCQ] In this model, what distinguishes a hub from an authority?

- A. A hub points outward to content (a playlist); an authority is pointed at (a song).
- B. A hub is pointed at by others; an authority points outward to other nodes.
- C. A hub and an authority are simply two names for the very same single score.
- D. A hub is any node with no out-links; an authority is any node with no in-links.

Answer: A

Reference: Lecture 104 – Hubs and Authorities (≈00:39–02:40)

Solution: A hub points outward (a playlist that features songs), and an authority is the thing pointed at (a song). The roles are reversed in one option, they are not the same number, and they are defined by pointing, not by having zero links.

Q2. [MCQ] Under the principle of repeated improvement, how are the two scores updated each round?

- A. A hub's score becomes the count of every node in the whole graph, and an authority's score likewise.

- B. Each score is replaced by its single largest neighbour score and then left fixed thereafter.
- C. A hub's score becomes the sum of authority scores it points to; an authority's, the hub scores pointing in.
- D. A hub's score becomes the sum of the hub scores it points to, and the authorities are ignored.

Answer: C

Reference: Lecture 102–103 – Principle of Repeated Improvement

Solution: Each hub takes on the total authority score of what it points to, and each authority takes on the total hub score of what points to it — the two roles feed each other. It is not a node count, not a single maximum, and it does mix hubs with authorities (not hubs with hubs).

Q3. [MCQ] Which statement best captures the recursive idea behind the scores?

- A. A song is a good authority purely because a large number of playlists feature it.
- B. A playlist is a good hub if it features many songs, however obscure those songs are.
- C. A song's score depends only on how many other songs it happens to resemble in style.
- D. A song scores well when good playlists feature it, and good playlists feature good songs.

Answer: D

Reference: Lecture 104 – Hubs and Authorities (≈00:39–end)

Solution: The quality of a song depends on the quality of the playlists featuring it, which in turn depends on the songs they feature — a mutually recursive definition. Raw counts of playlists or songs, and stylistic resemblance, miss this importance-weighted recursion.

Q4. [MCQ] In a fully general directed graph (not a clean playlist-vs-song split), how many scores does each node carry?

- A. One score only, decided by whether the node has more incoming links or more outgoing links.
- B. Two scores: a hub score for what it points to and an authority score for who points to it.
- C. One shared score that every node in the same community ends up holding in common.
- D. Three scores: a hub score, an authority score, and a separate node-degree score.

Answer: B

Reference: Lecture 104 – Hubs and Authorities (≈01:11–04:00)

Solution: In a general directed graph every node both points and is pointed at, so it carries two scores at once — a hub score and an authority score. It is not reduced to one score, not shared across a community, and there is no separate third degree score in this method.

Q5. [Match the following] Match each term in Column A with its description in Column B.

Column A (term)	Column B (description)
1. Hub	a. its score is the sum of the hub scores pointing to it
2. Authority	b. relative scores stop changing as the update repeats

3. Repeated improvement	c. its score is the sum of the authority scores it points to
4. Convergence	d. alternately update hub and authority scores, normalizing each round

- A. 1-a, 2-c, 3-d, 4-b
- B. 1-c, 2-a, 3-b, 4-d
- C. 1-c, 2-a, 3-d, 4-b
- D. 1-c, 2-d, 3-a, 4-b

Answer: C (1-c, 2-a, 3-d, 4-b)

Reference: Lectures 102–104 – Repeated improvement and convergence

Solution: A hub's score is the sum of the authority scores it points to (1-c); an authority's score is the sum of the hub scores pointing to it (2-a); repeated improvement alternately updates and normalizes the two score sets (3-d); and convergence is when the relative scores stop changing (4-b).

Q6. [MCQ] What does it mean for the hub and authority scores to “converge”?

- A. Every node ends up with exactly the same score as every other node in the graph.
- B. The relative proportions between nodes' scores stop changing as the update repeats.
- C. The raw scores grow without bound, doubling on every successive iteration.
- D. All hub scores fall to zero while the authority scores keep rising each round.

Answer: B

Reference: Lecture 104 – Hubs and Authorities (≈05:53–end)

Solution: Convergence here means the proportions between nodes settle — if one node holds half as much as another, it keeps doing so — even as the numbers are rescaled. The scores do not all become equal, do not blow up (that is what normalizing prevents), and hubs do not collapse to zero.

Q7. [MCQ] Why are the scores normalized after every round of updates?

- A. To force every node's score to land at exactly one, removing all differences.
- B. To keep the raw scores from growing without bound, since only proportions matter.
- C. To convert the directed graph into an undirected one before the next round.
- D. To discard the authority scores so that only the hub scores remain in play.

Answer: B

Reference: Lecture 103 – Principle of Repeated Improvement (An example)

Solution: Summing neighbour scores each round would make the numbers grow indefinitely, so normalizing rescales them while preserving the relative proportions that actually carry the ranking. It does not equalise the scores, change the graph's direction, or delete the authority scores.

Q8. [MCQ] Playlist X features songs P and Q; playlist Y features songs Q and R. Starting every score at 1, what is the (un-normalized) authority score of song Q after one update?

- A. 0, because Q is cancelled out by appearing on two different playlists.
- B. 1, because each authority score is capped at one before normalizing.
- C. 3, the number of distinct songs across both of the playlists combined.
- D. 2, since Q is featured by both X and Y, each contributing a hub score of 1.

Answer: D

Reference: Lecture 103 – Principle of Repeated Improvement (An example)

Solution: An authority's score is the sum of the hub scores pointing to it. Q is featured by both X and Y, each with hub score 1, so Q's authority score is $1 + 1 = 2$ (P and R get 1 each). It is not cancelled, not capped at one, and not the count of distinct songs.

Q9. [MCQ] For which graphs is this repeated-improvement process guaranteed to converge?

- A. For any directed graph — the hub and authority scores always reach a stable point.
- B. Only for graphs in which every hub points to exactly the same set of authorities.
- C. Only for undirected graphs, since direction would prevent the scores from settling.
- D. Only for graphs that contain no cycles of any kind among their nodes.

Answer: A

Reference: Lecture 104 – Hubs and Authorities (≈05:53–end)

Solution: The principle of repeated improvement is guaranteed to converge for any graph — the scores always reach a stable set of proportions. It does not require identical hubs, undirected edges, or acyclic structure.

Q10. [MSQ] Which statements about hubs and authorities are correct? (Select all that apply.)

- I. A hub's score is the sum of the authority scores of the nodes it points to.
 - II. In a general directed graph, each node has both a hub and an authority score.
 - III. The scores are normalized each round, and the proportions converge.
 - IV. A node's authority score depends only on its number of out-links.
- A. I, II, IV
 - B. II, III, IV
 - C. I, II, III
 - D. I, III

Answer: C (statements I, II, III)

Reference: Lectures 102–104 – Hubs and authorities

Solution: A hub aggregates the authority scores it points to (I), each node carries both scores (II), and the normalized scores converge in proportion (III). An authority score reflects who points in (in-links and their hub scores), not the node's out-links, so IV is false.

Comprehension 2 — PageRank Revisited: Resource-Sharing as a Matrix Operation

Case Study

Returning to PageRank, an instructor reframes it as a resource-sharing process on a directed graph and then as a matrix operation. Each of the n nodes starts with an equal share, $1/n$, of the total. In every round, a node hands its current value out along its links, split equally among the nodes it points to. Because nothing leaves the network, the values always sum to 1 a conservation law, like conservation of energy and after enough rounds the shares stop changing, so the process converges to a stable assignment.

A concrete graph on three nodes A, B, C has the links $A \rightarrow C$, $B \rightarrow A$, $B \rightarrow C$, and $C \rightarrow B$. Starting from an equal $1/3$ each and applying the sharing rule, one can follow how the shares shift from one round to the next and where they finally settle (remembering that B, with two out-links, splits its value between the two nodes it points to).

The instructor then shows this sharing step is exactly multiplication by a matrix M whose entry in row i , column j is the share that node j sends to node i . This M turns out to be a Markov matrix, and the iteration is just repeated multiplication of the share vector by M . A key fact from linear algebra makes the convergence inevitable: repeatedly applying a matrix to any starting vector drives it toward the eigenvector of the largest eigenvalue, and the special structure of a Markov matrix pins down a single limiting assignment, the same no matter where the shares start.

Questions 1–10

Q1. [MCQ] How does the resource-sharing version of PageRank begin, and what happens each round?

- A. Each node starts with 1 full point, and each round it simply keeps whatever its in-links bring.
- B. Each node starts with $1/n$ of the total and splits its value equally along its out-links.
- C. Each node starts with $1/n$ and passes all of its value to a single random neighbour.
- D. Only the highest-degree node starts with value, which then trickles out to the others.

Answer: B

Reference: Lecture 105 – PageRank Revisited: An example (≈01:09–03:24)

Solution: Every node begins with an equal share $1/n$, and each round it divides its current value equally among the nodes it points to. It does not start at 1, does not send everything to one random neighbour, and the value is spread evenly, not hoarded by one node.

Q2. [MCQ] Why do the node values always sum to 1 throughout the process?

- A. Because the values are re-scaled to sum to 1 by hand after every single round.
- B. Because each node's value is capped at $1/n$ so the total can never exceed one.
- C. Because the highest-value node is reset to absorb any shortfall in the total.
- D. Because value only moves along edges and never leaves the network — conservation.

Answer: D

Reference: Lecture 107 – Conservation and Convergence (≈00:05–end)

Solution: The shares are merely passed along edges, never created or destroyed, so the total is conserved at 1 — the instructor likens it to conservation of energy. No manual rescaling, per-node cap, or top-node reset is involved.

Q3. [MCQ] On the three-node graph ($A \rightarrow C$, $B \rightarrow A$, $B \rightarrow C$, $C \rightarrow B$), starting from $1/3$ each, what is the value at C after one round?

- A. $1/6$, the value handed from a single one of C's in-links.
- B. $1/3$, unchanged from the equal starting share each node was given.
- C. $1/2$, since C collects all of A's value plus half of B's value.
- D. $2/3$, the combined value of every node that B happens to point to.

Answer: C

Reference: Lecture 105 – PageRank Revisited: An example (≈03:24–end)

Solution: C receives all of A's $1/3$ (A points only to C) and half of B's $1/3$ (B splits between A and C), giving $1/3 + 1/6 = 1/2$. The other values are $1/6$ for A and $1/3$ for B that round.

Q4. [MCQ] The three-node process converges to $A = 0.2$, $B = 0.4$, $C = 0.4$. Which reading is correct?

- A. B and C end up ranked equally and ahead of A, and the three shares sum to 1.
- B. A ends up ranked highest, since it started by giving its value straight to C.
- C. The three shares converge but no longer add up to 1 once they have settled.
- D. All three nodes converge to one third each, exactly as they began the process.

Answer: A

Reference: Lecture 106 – Convergence in the Example (≈04:49–end)

Solution: The stable shares are $A = 0.2$, $B = 0.4$, $C = 0.4$: B and C tie for the top and A is lowest, and $0.2 + 0.4 + 0.4 = 1$ as conservation requires. A is not highest, the total stays 1, and the shares move away from the equal start.

Q5. [MCQ] Once the sharing step is written as multiplication by the matrix M, what is special about M?

- A. Every row of M sums to 1, which makes M a symmetric adjacency matrix.
- B. Each column of M sums to 1 — a column-stochastic (Markov) matrix.
- C. M has all entries equal, so every node sends the same share everywhere.
- D. M is its own inverse, so applying it twice returns the original vector.

Answer: B

Reference: Lecture 114 – PageRank Explained (≈03:09–04:36)

Solution: M's entry (i, j) is the share node j sends to node i, and since j hands out all of its value, each column sums to 1 — a column-stochastic, or Markov, matrix. Its columns (not rows) sum to 1, its entries are not all equal, and it is not its own inverse.

Q6. [MCQ] Why does repeatedly multiplying the share vector by the Markov matrix M converge to a unique result?

- A. Because every possible starting vector is already an eigenvector of M , so nothing ever changes.
- B. Because M shrinks every vector toward the origin until all of its entries reach zero.
- C. Because M 's largest eigenvalue is 1, so the iteration is driven to its eigenvalue-1 eigenvector.
- D. Because the result depends entirely on the starting vector, giving a different limit each time.

Answer: C

Reference: Lecture 114 – PageRank Explained (≈04:36–end)

Solution: Repeated multiplication drives any vector toward the eigenvector of the largest eigenvalue; for a Markov matrix that eigenvalue is exactly 1, so the shares converge to M 's eigenvector for eigenvalue 1 no matter where they start. Not every vector is an eigenvector, M does not collapse vectors to zero, and the limit does not depend on the start.

Q7. [Arrange in order] Arrange the steps of viewing PageRank as a matrix operation in the correct order.

Steps (in scrambled order):

- (i) Start from the uniform vector with $1/n$ in every entry.
 - (ii) Repeatedly multiply the current vector by M .
 - (iii) Build the Markov matrix M whose columns each sum to 1.
 - (iv) Read off the stable vector as the PageRank scores.
 - (v) Stop once the vector reaches M 's eigenvector for eigenvalue 1.
- A. iii → i → ii → v → iv
 - B. i → iii → ii → v → iv
 - C. iii → ii → i → v → iv
 - D. iii → i → v → ii → iv

Answer: A (iii → i → ii → v → iv)

Reference: Lectures 113–114 – PageRank as a matrix operation

Solution: Build the Markov matrix M with columns summing to 1 (iii), start from the uniform $1/n$ vector (i), repeatedly multiply by M (ii), stop when the vector reaches M 's eigenvector for eigenvalue 1 (v), and read off that stable vector as the scores (iv).

Q8. [MCQ] A Markov matrix has largest eigenvalue 1 and all others smaller. Why does that force convergence under repeated multiplication?

- A. In the eigenvector basis, the smaller eigenvalues' powers vanish, leaving only the eigenvalue-1 direction.
- B. Because a largest eigenvalue of 1 means every eigenvalue equals 1, so the vector never changes at all.
- C. Because the eigenvalues below 1 grow fastest under repeated powers and so take over the result.
- D. Because the matrix has no eigenvectors, so the iteration must settle on the average of all entries.

Answer: A

Reference: Lecture 112 – Convergence in Repeated Matrix Multiplication: The Details

Solution: Express the start as a combination of eigenvectors; raising M to a high power scales each component by its eigenvalue to that power. Components with eigenvalue below 1 vanish, leaving only the eigenvalue-1 direction. The other options misstate the eigenvalues or deny that M has eigenvectors.

Q9. [MCQ] The convergence argument rests on a prerequisite about adding vectors. What is it?

- A. Adding two vectors of the very same length together always yields exactly the zero vector.
- B. Adding any two of the vectors reverses both of their directions at the very same moment.
- C. The sum of two of the vectors is always perpendicular to each of the two originals.
- D. A big vector plus a small vector gives a result close to the big vector's direction.

Answer: D

Reference: Lecture 111 – Addition of Two Vectors (Prerequisite 2)

Solution: When a large-amplitude vector is added to a small one, the resultant points close to the large vector — which is why the dominant eigenvalue's term determines the limiting direction. Equal-length vectors need not cancel, addition does not reverse directions, and the sum is not generally perpendicular to the parts.

Q10. [MSQ] Which statements about PageRank as a matrix operation are correct? (Select all that apply.)

- I. The node values are conserved, summing to 1 throughout the process.
 - II. The sharing step is multiplication by a Markov matrix whose columns sum to 1.
 - III. The shares converge to the eigenvector for eigenvalue 1, regardless of the start.
 - IV. The final shares depend on the starting vector, so each run gives a different ranking.
- A. I, II, IV
 - B. II, III, IV
 - C. I, III, IV
 - D. I, II, III

Answer: D (statements I, II, III)

Reference: Lectures 107–114 – Conservation, convergence, and the matrix view

Solution: Conservation keeps the total at 1 (I), the step is multiplication by a column-stochastic Markov matrix (II), and the shares converge to the eigenvalue-1 eigenvector independent of the start (III). Because the limit is independent of the start, IV is false.